

To avoid being spoofed by hackers:

- The network administrator should deploy strict verification methods before accepting any IP packets from the hacker's computer. For instance, administrators can deploy authentication of packets based on a key exchange between two machines on the same network using IPsec.
- Do not allow any private IP addresses to access the network. Authenticate the IP address of every single packet that is inbound to the server and uses encryption protocols such as *HTTPS*.
- It is recommended for website developers to make use of the IPv6 protocol that has the inbuilt capability of both encryption and authentication of network packets.
- Have filtering in place both for inbound and outbound network traffic.

Auditing processes for prevention

It is always a best practice for large organizations to perform a detailed audit process of their computers. This helps them determine any security lapses beforehand and get them fixed before something goes wrong or the data integrity is compromised. As a process, it's better to run the audit process frequently to define and validate security policies in the organization. The audit process has to be performed on all the computers within the office network to make sure every system adheres to the security policies.

This section will list the different strategies to keep your computers and mobile phones safe and stay away from hackers.

1. **Software updates and patches** – Ensure all the software in the computers is running on the latest versions. Also, keep an eye on the security patches that are released by software vendors with updates to newly found issues. If there is any unwanted software or freeware in the system, the network administrator should immediately remove them from the machines as they can be the perfect set-up for hackers to get into the system.
2. **Two-Factor Authentication** – The best way to ensure full security for the websites is to deploy two-factor authentication to allow only authorized access to the web application or web server. When too many login attempts are detected, a message is automatically sent to the connected mobile number with a temporary verification code. This helps to add an extra layer of security for the data.
3. **Password Strength** – Ensure that you use strong passwords with a combination of alphanumeric characters and special characters. Don't use common terms like “password, welcome@123, India, ...” as passwords